OJT-1 Report

Job Role: AI Engineer Intern

During the second phase of my internship at **TestHeapStack**, I worked as an AI Engineer Intern. My primary assignment was the design and end-to-end development of the **"Fabricated News Prediction System"** — an AI-powered platform designed to detect and classify fake news articles using Natural Language Processing (NLP) and Machine Learning techniques.

This project aligns with TestHeapStack's mission of building intelligent, AI-driven solutions—applied here to the critical domain of misinformation detection and digital content authenticity.

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1. Application of Skills and Technical Knowledge

Throughout this internship phase, I applied a diverse set of technical skills across AI/ML, backend development, and frontend integration for the Fabricated News Prediction System.

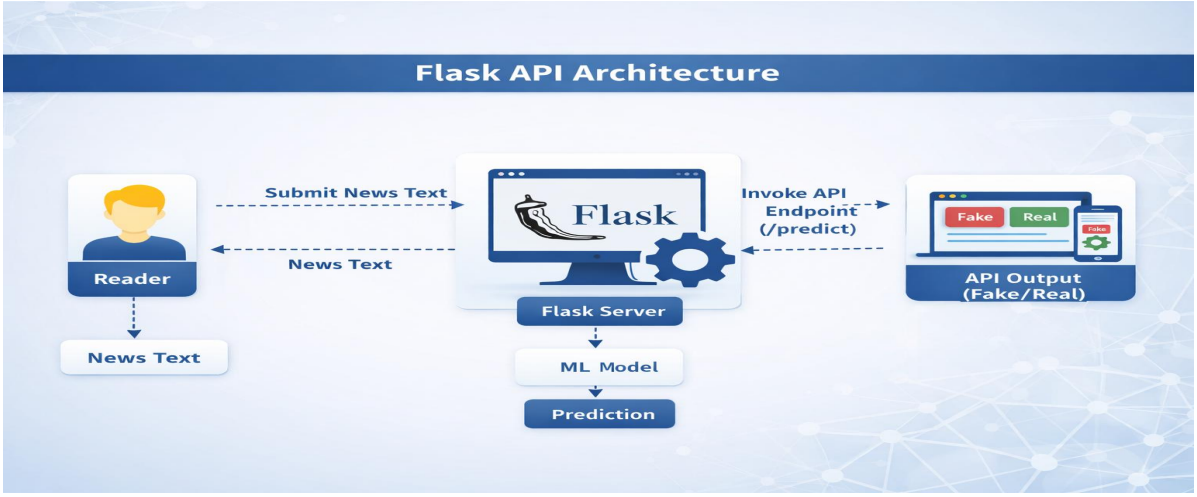
Python served as the primary programming language across all components—from data preprocessing and NLP-based feature extraction to model training and deployment through a web interface.

AI/ML Stack:

- **Predictive Modeling:**
  Used Scikit-learn to build classification models such as Logistic Regression and Naïve Bayes for fake news detection. The trained model was serialized using Joblib for efficient deployment.
- **Natural Language Processing (NLP):**
  Applied TF-IDF Vectorization for transforming textual news data into numerical features, enabling effective classification.
- **Data Engineering:**
  Utilized Pandas and NumPy for cleaning, preprocessing, and structuring large datasets of news articles.
- **Model Integration:**
  Integrated the trained ML model with the Flask backend to enable real-time prediction of news authenticity

Backend Development

- Developed a RESTful backend using Flask.
- Handled user inputs (news text) and returned classification results (Real/Fake).
- Integrated model inference with API endpoints.



Frontend Development

- Built an interactive UI using Streamlit for quick deployment and testing.
- Designed a clean interface for users to input news content and view prediction results.
- Ensured responsive and user-friendly experience

2. Performance on Assigned Tasks and Projects

Implementation Completed

- Developed a complete pipeline for fake news detection.
- Implemented preprocessing, feature extraction, model training, and deployment.
- Created an interactive web application for real-time predictions.

Key Achievements:

- Successfully trained models on large datasets with good accuracy.
- Built an end-to-end system from data ingestion to prediction.
- Achieved seamless integration between frontend and backend.

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3. Value Addition to the Organization

- **Combatting Misinformation:**
  Helps identify fake news and reduces the spread of misinformation.
- **Automation:**
  Automates manual verification of news content using AI.
- **Scalable Solution:**
  Modular design allows easy integration of advanced NLP models in future.

- **User Awareness:**
  Provides users with quick insights into the authenticity of news articles.

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4. Job Role Understanding and Relevance

The role of AI Engineer Intern at TestHeapStack for the Fabricated News Prediction System required:

- Implementing NLP techniques for text classification.
- Designing efficient feature extraction methods like TF-IDF.
- Building and deploying machine learning models.
- Developing backend APIs using Flask.
- Creating interactive user interfaces for real-time predictions.

This role provided hands-on experience across the full lifecycle of an AI application—from raw data preprocessing to deploying a working prediction system.

System Architecture Overview

- **Frontend:** Streamlit
- **Backend:** Flask
- **AI Engine:** Scikit-learn + TF-IDF
- **Database:** (Optional / Dataset-based system)
- **Architecture:** Modular and scalable ML pipeline

Use Case Documentation

Use Case Title: Fake News Detection Intelligence Description:

The system accepts news articles or text input, processes them using NLP techniques, and classifies them as **Real or Fake** using a trained machine learning model.

Real-World Example:

A user enters a suspicious news headline or article into the system and asks,
"Is this news real or fake?"
The system analyzes the text and instantly returns a classification result with confidence.

Actors:

- User (Reader / Journalist / Researcher)
- Fabricated News Prediction System

Inputs:

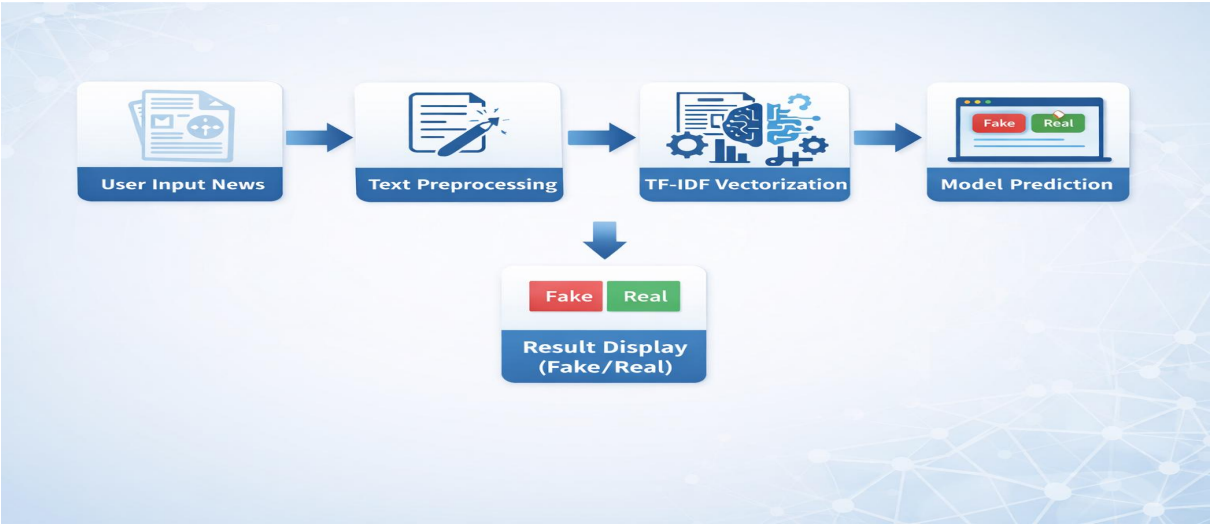
- News article text
- User query/input

Process:

- Text Input: User enters news content.
- Preprocessing: Cleaning, tokenization, stopword removal.
- Feature Extraction: TF-IDF vectorization.
- Prediction: ML model classifies the news.
- Output Generation: Displays result (Fake/Real).

Workflow:

```
[User Input News]
      ↓
[Text Preprocessing]
      ↓
[TF-IDF Vectorization]
      ↓
[Model Prediction]
      ↓
[Result Display (Fake/Real)]
```



Output:

- Prediction: Fake or Real
- Confidence Score

Advantages:

- Fast and automated fake news detection
- Reduces spread of misinformation
- Easy-to-use interface
- Scalable and extendable system

Limitations:

- Accuracy depends on training dataset quality
- Cannot detect sarcasm or highly nuanced content
- Requires regular model updates

Future Enhancements:

- Integration with deep learning models (LSTM, BERT)
- Real-time news API integration
- Multilingual fake news detection
- Browser extension for live news verification

Conclusion

During this internship at **TestHeapStack**, I successfully designed and developed the **Fabricated News Prediction System**, a complete AI-based solution for detecting misinformation.

This project provided practical exposure to machine learning pipelines, NLP techniques, and full-stack deployment. It demonstrates how AI can be effectively used to tackle real-world challenges like fake news detection and digital content verification.